

The Utility Problem

With the ever-increasing presence of climate change, extreme weather events featuring extreme temperatures are occurring more frequently. In addition, with the increasing presence of wildfires over recent years, grid assets can experience much higher temperatures. Every grid asset — from overhead conductors to pad-mounted transformers — has defined thermal ratings beyond which it operates inefficiently or risks failure. When thermal ratings are exceeded — thresholds that vary by asset type — the consequences include inefficient operation, unplanned outages, damage to adjacent assets, and, in the worst cases, wildfires.

Grid equipment is engineered to operate within defined thermal limits. When those limits are exceeded, efficiency drops sharply. This erodes utility revenue and increases emissions, as more generation capacity is needed to deliver the same load. Sustained overtemperature conditions can cause premature asset failure. A failed asset can trigger customer outages, service interruptions, and significant financial losses. Cascading failures can damage adjacent equipment or ignite wildfires through electrical arcing.

Overall, the impacts of extreme temperature on grid components can range from financial losses, increased emissions and customer dissatisfaction to wildfires, extended outages and other damaged assets. With this in mind, thermal monitoring of grid assets is increasingly critical as climate-driven heat events and wildfire risks escalate. Many utilities are deploying temperature-sensing IoT devices, providing valuable situational awareness — but correlating sensor readings with asset thermal ratings, weather forecasts, and GIS locations remains a largely manual process. Automating this correlation is where the most significant operational value lies.

The Data Solutions and Methods

This analysis draws on three key datasets:

1. Weather data corresponding to the desired locations on the distribution grid.
2. IoT sensor temperature readings.
3. Grid Asset specification data, including the operating temperature range.

This analysis can be completed efficiently in both our front-end application, True Grid Intelligence (TGI), and with Awesense Energy Data Model (EDM), accessible through our SQL EDM API. The Data Engine's comprehensive data model and built-in access functions make this analysis straightforward to implement. The analyst queries weather data and available IoT sensor temperature readings for the service territory via the EDM API. They then retrieve the thermal rating for each grid component. The analysis can then be automated to query hourly for temperature and weather data, and compare the temperature at the GIS location with the temperature at each grid component and its corresponding rating. Automated alerts notify the operations team when an asset's thermal rating is at risk of being exceeded. These outputs give

Grid Temperature Monitoring

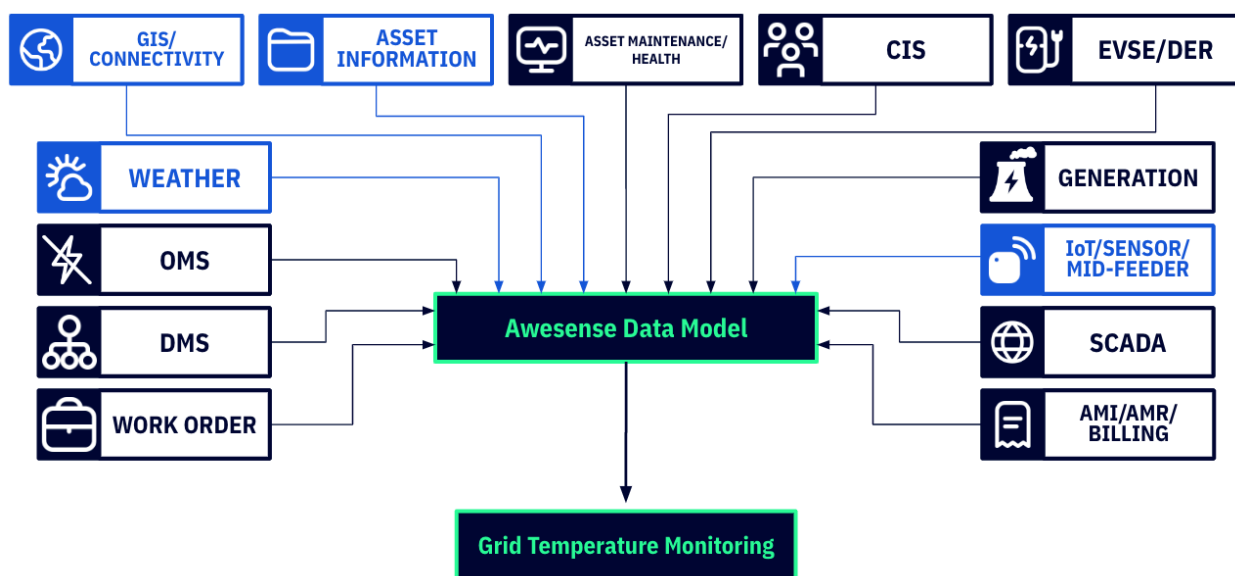
[UC12] - Awesense Platform Use Case



operations and planning teams the intelligence needed to adjust switching schedules, dispatch maintenance crews, and prioritize component upgrades before assets fail.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. This use case is implemented in TGI as part of the Grid Alerts functionality. Please contact [Awesense](#) if you would like to see a demo of this use case.